FACTORING THROUGH A SURJECTIVE FLAT COVER DETECTED ON A SCHEME-THEORETICALLY DENSE OPEN

THE STACKS PROJECT AUTHORS

Editorial note. No omission markers in-span.

Allowed standard prerequisites (editorial).

- Sheaf quotient by a flat covering. the identification Z = X/R as fppf sheaves for g surjective flat (structural use). Turns the covering into a groupoid presentation.
- Scheme-theoretic density under flat pullback. Morphisms of Spaces, lemma flat-morphism-scheme-theoretically-dense-open (cross-chapter, named). Pulls the dense open V back to a dense open in R.
- Equality of morphisms. Morphisms of Spaces, lemma equality-of-morphisms (cross-chapter, named). Two morphisms agreeing on a scheme-theoretically dense open are equal.

Source reference. Original reference 084N: focused-seeds/geometry/S9-smg-factor-through-over-open/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 089D: focused-seeds/geometry/S9-smg-factor-through-over-open/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let B be an algebraic space over S. Let $f : X \to Y$ and $g : X \to Z$ be morphisms of algebraic spaces over B. Assume*

(1) *$Y \to B$ is separated,*
(2) *g is surjective, flat, and locally of finite presentation,*
(3) *there is a scheme theoretically dense open $V \subset Z$ such that $f|_{g^{-1}(V)} : g^{-1}(V) \to Y$ factors through V.*

Then f factors through g.

Proof. Set $R = X \times_Z X$. By (2) we see that $Z = X/R$ as sheaves. Also (2) implies that the inverse image of V in R is scheme theoretically dense in R (Morphisms of Spaces, Lemma 089D). The we see that the two compositions $R \to X \to Y$ are equal by Morphisms of Spaces, Lemma 084N. The lemma follows. □

Date: stacks-project:spaces-more-groupoids.tex:local-snapshot-20261012; GNU FDL 1.2 or later.